Undermining Deterrence

The Case for a Total Ban on DoD Research Involving Chinese Military Companies

AUTHOR
LJ Eads

APRIL 2025

Parallax
ADVANCED RESEARCH

Executive Summary

Despite clear congressional intent outlined in Section 1260H of the 2021 National Defense Authorization Act (NDAA), U.S. Department of Defense (DoD) funding continues to flow to Chinese Military Companies (CMCs)—entities formally designated by the U.S. government as threats to national security. This persistent failure of oversight has enabled at least $48 million in DoD-funded research to directly or indirectly benefit Chinese military-linked firms, including Huawei, BGI, China General Nuclear Power Group, and China Mobile—core components of China's military-industrial complex. Although current DoD policy requires mitigation for affiliations with entities listed under Section 1260H of the FY2021 NDAA and Section 1286 of the FY2019 NDAA, these measures have proven insufficient. Mitigation protocols have failed to stop adversarial access to sensitive U.S. defense research. Mitigation measures are discretionary, informal, and inconsistently applied. While the law identifies risk, it does not mandate how that risk must be addressed—leaving enforcement up to individual program officers and research institutions, with no mandatory prohibition mechanism in place. The continued presence of Chinese Military Companies in DoD-funded academic collaborations reveals not only the inadequacy of existing safeguards—but the urgent need for a categorical prohibition on all such engagements.

Findings: U.S. DoD Funds Supporting Chinese Military Companies

A review of federally funded academic and scientific research has identified seventeen specific cases in which Chinese Military Companies participated in or benefited from DoD-funded research. These cases involve top U.S. research institutions—including the Massachusetts Institute of Technology (MIT), Virginia Polytechnic Institute & State University, University of California Santa Cruz, Washington University, Vanderbilt University, Yale University, and Princeton University—engaged in partnerships with Chinese military-linked firms on projects directly funded by the DoD. DoD's current policy guidance—codified in its internal decision matrix—identifies affiliations with Section 1260H and Section 1286 entities as research security risks requiring mitigation. However, this approach places the burden on individual program officers and research institutions to interpret risk, rather than enforcing a clear prohibition. This has led to inconsistent application, delayed interventions, and continued access by adversarial entities.

Key examples include:

- MIT's Army Research Office-funded project ($24M) involving Huawei Technologies (China), a known Chinese state-controlled telecommunications and AI company with extensive ties to the People's Liberation Army (PLA).
- Virginia Polytechnic Institute's ONR-funded study ($3.5M) collaborating with Huawei Technologies (France), exposing U.S. defense-funded cybersecurity, computing, and AI research to a company that has been repeatedly flagged for espionage and state-sponsored cyber activities.
- University of California Santa Cruz's ONR-funded research ($547K) featuring BGI Research (China), a company linked to China's state-led genetic surveillance programs and human genome research for military applications.
- Washington University's Army-funded study ($1.45M) involving China Mobile Group Co., Ltd., a PLA-affiliated entity in China's telecommunications intelligence network.
- Vanderbilt University's Air Force-funded research ($1.4M) with Huawei Technologies (China), further demonstrating the systemic failures in restricting access to U.S. military-funded research.

Each of these cases demonstrates how ineffective oversight has allowed Chinese Military Companies to infiltrate the U.S. research ecosystem, leveraging American defense dollars to enhance their own technological capabilities in semiconductors, quantum computing, artificial intelligence, cyber warfare, telecommunications, and biotechnology—all strategic priority areas for China's military modernization.

Introduction

Despite clear congressional intent under Section 1260H of the 2021 National Defense Authorization Act (NDAA) [1], the U.S. Department of Defense (DoD) continues to fund research collaborations involving Chinese Military Companies (CMCs)—entities officially designated as threats to U.S. national security. This ongoing entanglement is not merely an oversight—it reflects a deeper failure in enforcement, where discretionary mitigation has replaced mandatory exclusion.

Current DoD policy recognizes the risks posed by affiliations with Chinese military-linked entities, and it requires mitigation under internal research security guidance. However, mitigation measures are informal, inconsistently applied, and ultimately insufficient. The law identifies the threat but does not mandate how to prevent it—leaving implementation up to individual program officers and institutions, without a formal prohibition mechanism.

A recent review of federally funded academic and scientific collaborations has uncovered at least $48 million in DoD grants that directly or indirectly benefited Chinese Military Companies, including Huawei, BGI, China General Nuclear Power Group, and China Mobile. These firms are central to China's military-industrial complex and have documented ties to the People's Liberation Army (PLA). The collaborations span advanced fields such as artificial intelligence, telecommunications, quantum computing, and biotechnology—technologies prioritized in China's military-civil fusion strategy.

This failure in oversight presents an immediate and escalating risk to U.S. defense capabilities and technological superiority. Unless policymakers move beyond mitigation and enforce a strict, categorical prohibition on DoD-funded collaborations with Chinese Military Companies, the United States will continue to strengthen its most capable adversary—at the direct expense of national security.

DoD-Funded Research Collaborations with Chinese Military Companies

1. **Designing Structures that Maximize Spatially Averaged Surface-Enhanced Raman Spectra** (November 2024) – This study, funded by the Army Research Office under a $24M cooperative agreement (W911NF-18-2-0048) with

1 DOD Releases List of Chinese Military Companies in Accordance with Section 1260H of the National Defense Authorization Act for Fiscal Year 2021, Jan. 7, 2025, https://www.defense.gov/News/Releases/Release/Article/4023145/dod-releases-list-of-chinese-military-companies-in-accordance-with-section-1260/

Massachusetts Institute of Technology, involved Huawei Technologies (China) as a collaborator.[2]

2. **HEXO: Offloading Long-Running Compute- and Memory-Intensive Workloads** (October 2024) – Funded by the Office of Naval Research through Virginia Polytechnic Institute & State University, this study featured collaboration with Huawei Technologies (France). The project received $723K (N00014-16-1-2104), $650K (N00014-16-1-2711), $1.9M (N00014-19-1-2493), and $241K (N00014-22-1-2672) in DoD funding. The U.S. researchers were also funded by the National Science Foundation.[3]

3. **Spatial Bayesian Neural Networks** (April 2024) – Conducted at The University of Wollongong, this project received $299K (FA2386-23-1-4100) in DoD funding from the Air Force Office of Scientific Research and included Huawei Technologies (France) as a research partner.[4]

4. **Genomics of Post-Bottleneck Recovery in the Northern Elephant Seal** (February 2024) – BGI Research (China) participated in this study, which was funded through the Office of Naval Research at University of California, Santa Cruz, with $547K (N00014-18-1-2822) awarded, while an additional amount under N000014-13-1-0134 remains unspecified. The U.S. researchers were also funded by the National Science Foundation.[5]

5. **Order Transfer in a Hybrid Raman-Laser-Optomechanical Resonator** (October 2023) – China Mobile Group Co., Ltd. collaborated on this study, which received DoD funding from the Army Research Office at Washington University, with $1M (W911NF-12-1-0026) and $450K (W911NF-17-1-0189) in awarded funds. The U.S. researchers were also funded by the National Science Foundation.[6]

---

2 Ian M. Hammond, Pengning Chao, Wenjie Yao, Francesc Verdugo, Henry O. Everitt, Rasmus E. Christiansen, and Steven G. Johnson, "Designing structures that maximize spatially averaged surface-enhanced Raman spectra: erratum," Opt. Express 32, 44754-44755 (2024), https://opg.optica.org/oe/fulltext.cfm?uri=oe-32-25-44754&id=564631

3 P. Olivier et al., "HEXO: Offloading Long-Running Compute- and Memory-Intensive Workloads on Low-Cost, Low-Power Embedded Systems" in IEEE Transactions on Cloud Computing, vol. 12, no. 04, pp. 1415-1432, Oct.-Dec. 2024, doi: 10.1109/TCC.2024.3482178., https://www.computer.org/csdl/journal/cc/2024/04/10720200/215PAl5fP68

4 Zammit-Mangion, A., Kaminski, M. D., Tran, B.-H., & Filippone, M. (2024). Spatial Bayesian neural networks. Spatial Statistics, 60, 100825., https://www.sciencedirect.com/science/article/pii/S2211675324000162?via%3Dihub

5 Hoelzel, A.R., Gkafas, G.A., Kang, H. et al. Genomics of post-bottleneck recovery in the northern elephant seal. Nat Ecol Evol 8, 686–694 (2024)., https://www.nature.com/articles/s41559-024-02337-4

6 Zhenning Yang, Wanzhe Zhang, Changlong Zhu, Zhe Wang, Jianqing Guan, Yue Huo, Xiaohe Tang, Wei Shi, Keyu Xia, Yu-xi Liu, Lan Yang, and Jing Zhang, "Order transfer in a hybrid Raman-laser-optomechanical resonator," Opt. Express 31, 36836-36844 (2023), https://opg.optica.org/oe/fulltext.cfm?uri=oe-31-22-36836&id=540907

6. **Automating Recoverability Proofs for Cyber-Physical Systems** (June 2023) – Conducted at SRI International, this research included Huawei Munich Research Center and received funding from the Office of Naval Research, totaling $1M (N00014-15-1-2202) and $540K (N00014-20-1-2644). Additional funding was involved from the Naval Research Laboratory (N0017317-1-G002) to University of Illinois for $925K.[7]

7. **Low-Frequency and Random Telegraph Noise in 14-nm Bulk Si Charge-Trap Transistors** (April 2023) – Huawei Technologies (China) was involved in this study, which received $1.4M (FA9550-17-1-0046) from the Air Force Office of Scientific Research at Vanderbilt University.[8]

8. **Video Joint Denoising and Demosaicing with Recurrent CNNs** (January 2023) – Conducted at École Normale Supérieure Paris-Saclay, this study, involving Huawei Technologies (France), was supported by the Office of Naval Research under $300K (N00014-17-1-2552) in funding.[9]

9. **Age-Aware Stochastic Hybrid Systems** (January 2023) – This research involved Huawei Technologies (France) and received DoD funding through two institutions: The Ohio State University, awarded $782K (N00014-17-1-2417), and University of Maryland, College Park, awarded $299K (N00014-18-1-2046) under the Office of Naval Research. The U.S. researchers were also funded by the National Science Foundation.[10]

DoD-Funded Research Co-Funded by Chinese Military Companies

10. **Privacy Preserving Semi-Decentralized Mean Estimation Over Intermittently-Connected Networks** (October 2024) – Researchers from Princeton University received DoD funding from an Air Force Office of Scientific Research award of an unknown amount (002484665) and additionally were funded in part by a Huawei Intelligent Spectrum Grant. The U.S. researchers were also funded by the National Science Foundation.[11]

---

7 Nigam, V., & Talcott, C. (2023). Automating recoverability proofs for cyber-physical systems with runtime assurance architectures [Technical report]. arXiv. https://arxiv.org/abs/2304.11564
8 M. Gorchichko et al., "Low-Frequency and Random Telegraph Noise in 14-nm Bulk Si Charge-Trap Transistors," in IEEE Transactions on Electron Devices, vol. 70, no. 6, pp. 3215-3222, June 2023, doi: 10.1109/TED.2023.3265939., https://ieeexplore.ieee.org/document/10108908
9 V. Dewil et al., "Video joint denoising and demosaicing with recurrent CNNs," 2023 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), Waikoloa, HI, USA, 2023, pp. 5097-5108, doi: 10.1109/WACV56688.2023.00508., https://ieeexplore.ieee.org/document/10030104
10 Maatouk, A., Assaad, M., & Ephremides, A. (2021). Age-aware stochastic hybrid systems: Stability, solutions, and applications [Preprint]. arXiv. https://arxiv.org/abs/2109.03919
11 R. Saha, M. Seif, M. Yemini, A. J. Goldsmith and H. V. Poor, "Privacy Preserving Semi-Decentralized Mean Estimation Over Intermittently-Connected Networks," in IEEE Transactions on Signal Processing, vol. 72, pp. 5306-5321, 2024, doi: 10.1109/TSP.2024.3473939., https://arxiv.org/pdf/2406.03766

11. **Collaborative Inference over Wireless Channels with Feature Differential Privacy** (October 2024) – Once more, researchers from Princeton University received DoD funding from an Air Force Office of Scientific Research award of an unknown amount (002484665) and additionally were funded in part by a Huawei Intelligent Spectrum Grant. The U.S. researchers were also funded by the National Science Foundation.[12]

12. **The columnar to equiaxed transition of CoCrNi medium-entropy alloy fabricated by laser directed energy deposition** (January 2024) **–** University of Tennessee at Knoxville researchers funded by Army Research Office awards totaling $646K (W911NF-13-1-0438 and W911NF-19-2-0049) collaborated with Harbin Institute of Technology researchers funded by the China General Nuclear Power Group (CGN) - Harbin Institute of Technology Advanced Nuclear and New Energy Research Institute. CGN has been sanctioned by the United States because CGN has attempted to acquire advanced U.S. nuclear technology to divert to military uses in China. The U.S. researchers were also funded by the National Science Foundation.[13] [14]

13. **Robust Semi-Decentralized Federated Learning via Collaborative Relaying** (December 2023) – Princeton University and Stanford University researchers received DoD funding from an Air Force Office of Scientific Research award of an unknown amount (002484665) and additionally were funded in part by a Huawei Intelligent Spectrum Grant.[15]

14. **UAV-Aided RF Mapping for Sensing and Connectivity in Wireless Networks** (August 2023) – A University of Southern California researcher supported by Office of Naval Research Grants totaling nearly $1.5M (N00014-15-1-2550) and an Army Research Office Grant for $9.7M (W911NF1910269) collaborated with EURECOM and Chinese University of Hong Kong researchers. The EURECOM researchers claimed funding via Huawei. The U.S. researchers were also funded by the National Science Foundation.[16]

---

12 Seif, M., Nie, Y., Goldsmith, A. J., & Poor, H. V. (2024). Collaborative inference over wireless channels with feature differential privacy (arXiv:2410.19917v1). arXiv. https://arxiv.org/abs/2410.19917
13 Zhao, W., Sun, Y., Che, P., Ning, Z., Fan, H., Yang, H., Sun, J., Liaw, P. K., Ngan, A. H. W., & Huang, Y. (2024). The columnar to equiaxed transition of CoCrNi medium-entropy alloy fabricated by laser directed energy deposition. Materials & Design, 237, 112538. https://doi.org/10.1016/j.matdes.2023.112538
14 Four China Nuclear Industry Companies Added to "Entity List", https://www.pillsburylaw.com/en/news-and-insights/china-industry-entity-list.html
15 Yemini, M., Saha, R., Ozfatura, E., Gündüz, D., & Goldsmith, A. J. (2023). Robust semi-decentralized federated learning via collaborative relaying. IEEE Transactions on Wireless Communications, 23(7), 7520–7536. https://doi.org/10.1109/twc.2023.3342095
16 Gesbert, D., Esrafilian, O., Chen, J., Gangula, R., & Mitra, U. (2022). UAV-aided RF mapping for sensing and connectivity in wireless networks (arXiv:2205.03335v1 [cs.IT]). arXiv. https://arxiv.org/pdf/2205.03335

15. **Collaborative Mean Estimation over Intermittently Connected Networks with Peer-To-Peer Privacy** (June 2023) - Princeton University and Stanford University researchers received DoD funding from an Air Force Office of Scientific Research award of an unknown amount (002484665) and additionally were funded in part by a Huawei Intelligent Spectrum Grant. The U.S. researchers were also funded by the National Science Foundation.[17]

16. **Adversarial Attacks for Black-Box Recommender Systems via Copying Transferable Cross-Domain User Profiles** (May 2023) – Michigan State University, other U.S. universities, and Hong Kong based universities acknowledged support by an Army Research Office Grant awarded $465K (W911NF-21-1-0198) and funding support from Huawei Research Collaborative Project and the Huawei Innovation Research Program. This DoD award was funded to The Pennsylvania State University and Michigan State University. The U.S. researchers were also funded by the National Science Foundation.[18] [19]

17. **Age Optimal Sampling Under Unknown Delay Statistics** (January 2023) **–** Yale University researchers leveraged an Office of Naval Research Grant totaling $1.5M (N00014-19-1-2566) directed to University of Illinois. The Chinese co-researchers from Tsinghua University received support from the Tsinghua University-China Mobile Research Institute Joint Innovation Center. The U.S. researchers were also funded by the National Science Foundation.[20]

Risks of Adversarial Technology Transfer Through DoD-Funded Research Collaborations

The entanglement of DoD-funded research with Chinese military-affiliated entities raises urgent national security concerns, particularly as adversaries like China seek to exploit emerging technologies for strategic advantage. For instance, research on UAV-aided RF mapping—a technology with clear military applications in electromagnetic sensing and precision targeting—was supported by over $11 million in Army and Navy grants, while involving researchers linked to Huawei and Chinese universities. Similarly, studies on

---

17 R. Saha, M. Seif, M. Yemini, A. J. Goldsmith and H. Vincent Poor, "Collaborative Mean Estimation over Intermittently Connected Networks with Peer-To-Peer Privacy," 2023 IEEE International Symposium on Information Theory (ISIT), Taipei, Taiwan, 2023, pp. 174-179, doi: 10.1109/ISIT54713.2023.10206910.
18 W. Fan et al., "Adversarial Attacks for Black-Box Recommender Systems via Copying Transferable Cross-Domain User Profiles," in IEEE Transactions on Knowledge and Data Engineering, vol. 35, no. 12, pp. 12415-12429, 1 Dec. 2023, doi: 10.1109/TKDE.2023.3272652. https://ieeexplore.ieee.org/document/10114977
19 Toward Interpretable and Stable Graph Neural Networks, Project Team Members, https://suhangwang.ist.psu.edu/projects/interpretable_robust_GNNs.html
20 Tang, H., Chen, Y., Wang, J., Yang, P., & Tassiulas, L. (2023). Age optimal sampling under unknown delay. arXiv. https://arxiv.org/pdf/2202.13367

federated learning, cyber-physical system recoverability, and spatial Bayesian neural networks, all critical to the future of autonomous systems, AI-enabled decision-making, and resilient communications, were carried out in coordination with Huawei's research labs in France and Germany. Even biotechnological research like the genomics study with BGI, and materials science projects involving CGN-funded Chinese researchers, highlight the strategic risks of dual-use knowledge transfer. These collaborations, often obscured within joint publications or secondary acknowledgments, create backdoor pipelines by which adversaries can accelerate military innovation using U.S. taxpayer resources. When American institutions partner—knowingly or unknowingly—with entities identified under Section 1260H of the NDAA, it undermines the integrity of the U.S. research enterprise and may directly enable China's military-civil fusion agenda to neutralize U.S. technological advantages across domains.

Policy Recommendations: From Mitigation to Prohibition

To prevent further entanglement of U.S. defense research with Chinese Military Companies (CMCs), immediate policy action is required to move beyond risk mitigation and toward categorical prohibition. The DoD's existing mitigation framework, as outlined in its Fundamental Research Proposal Decision Matrix, identifies affiliations with 1260H and 1286-listed entities as requiring heightened review. However, this approach has proven insufficient. The following recommendations are necessary to close persistent national security loopholes:

1. **Mandate a Categorical Ban on DoD Funding to Projects Involving CMCs**
   Congress should amend and enforce Section 1260H of the FY2021 NDAA to prohibit all DoD-funded research involving Chinese Military Companies, regardless of affiliation type (direct, indirect, or co-authorship). Any proposing institution found in violation should face funding disqualification across all DoD research accounts.

2. **Replace Risk-Based Mitigation with Automated Pre-Award Screening**
   DoD must implement a pre-award compliance system that automatically screens covered individuals and institutions against the 1260H and 1286 lists, as well as the BIS Entity List and EO 14032. Mitigation is not a viable substitute for preclusion in cases involving known PLA-linked entities.

3. **Strengthen Institutional Disclosure and Certification Requirements**
   All DoD-funded research institutions should be required to publicly disclose foreign research partnerships, funding sources, subawards, and co-authorships.

Annual certifications must confirm that no covered individuals or affiliated collaborators are tied to entities listed under Section 1260H or 1286.

4. **Establish a DoD Office for Research Security Enforcement**
   The DoD should establish a dedicated Research Security Enforcement Office with investigative authority, auditing capability, and the power to suspend awards or bar institutions from DoD funding in cases of noncompliance. This office should also liaise with the Intelligence Community to monitor foreign influence activity within academia.

5. **Expand Entity Designations and Sanctions for Research Exploitation**
   The Department of Commerce and Treasury should expand the BIS Entity List, SDN List, and EO 14032 Annex to include academic research programs, labs, and PLA-affiliated think tanks that co-publish or co-develop dual-use technologies with U.S. institutions. These designations must extend to foreign researchers operating under malign foreign talent recruitment programs defined in Section 1286(c)(9).

Conclusion: Enforcement, Not Mitigation, Is Now a National Imperative

The continued flow of DoD research funding to Chinese Military Companies—despite statutory restrictions under Section 1260H and clear national security guidance—reflects a systemic failure in oversight, not a lack of policy. Mitigation strategies have proven inadequate. By allowing adversarial entities linked to the PLA to access cutting-edge U.S. research in AI, biotechnology, and telecommunications, the U.S. is inadvertently accelerating China's military modernization and eroding its own technological edge.

This is not a policy gap—it is an enforcement crisis. Without immediate, categorical prohibitions on DoD-funded collaborations involving 1260H-listed entities, the United States will continue to subsidize the rise of its most capable strategic adversary.

Congress, the Department of Defense, and U.S. research institutions must act with urgency. Strengthening enforcement mechanisms, mandating full disclosure, and cutting off research ties to PLA-linked entities are not optional—they are essential to national security.

The time for risk-based mitigation has passed. The time for zero tolerance is now.